UI-Based Cryptography Exercises

1. Hashing with QuickHash

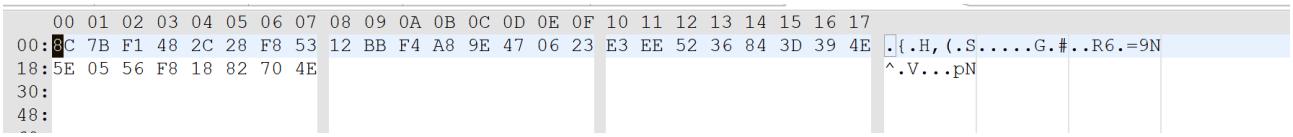
SHA3-256:
C49F824A42BD728D6A943A4B0ED1B0E7876E6F62D357E308AC29906B643C1A62

SHA3-512:
FE5F28D1896F9203626F271284337F5E6E32780FABD140BA206BBD08A4191F6F4B47CCCDE2015375CC8749C2ED81872940397586932E99C620CE687A821A1660

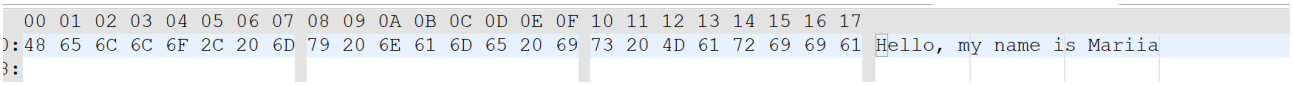
It provides a higher level of security because there are more possible combinations, making it mathematically harder to "guess" or find a collision

2. Encryption with JCrypTool

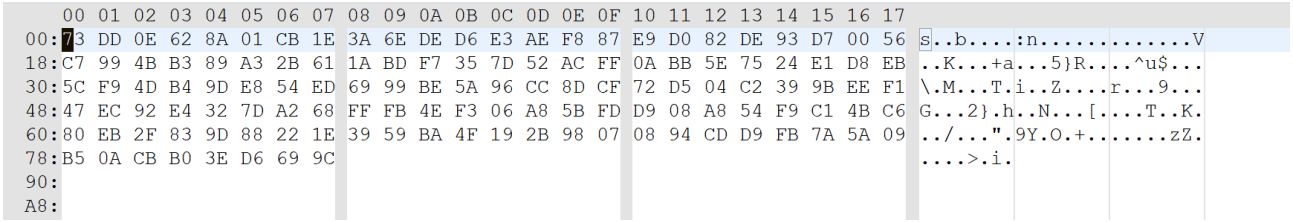
Symmetric encryption :



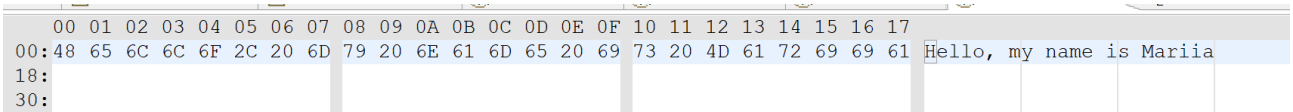
Decryption:



Asymmetric encryption :



Decryption:



Reflection:

Using JCrypTool and QuickHash made these concepts way easier to understand than just reading about them. In QuickHash, I saw the "Avalanche Effect" even changing one letter in my file completely changed the SHA-3 hash. JCrypTool also helped me visualize the difference between Symmetric and Asymmetric encryption by letting me actually manage the keys and watch my text turn into hex code.